G.M.M MIFTAHUL ALAM ADIB

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EDUCATION

B.Sc. in Statistics Nov 2024 – Present
Shahjalal University of Science and Technology, Sylhet

EXPERIENCE

Research Associate Aug 2025 – Present
Research Den Remote, United States
An AI/ML volunteer research community helping members publish in peer-reviewed conferences and journals through mentorship and collaboration (researchden.org).

- Co-authored a research manuscript currently under review, contributing across data analysis, experimentation, and writing.
- Active contributor across multiple ongoing research projects, applying machine learning techniques across the full research pipeline.

SKILLS

Programming Languages	Python, SQL
Databases	PostgreSQL, MySQL, SQLite
Machine Learning	PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLO, Hugging Face, Whisper, LangChain, FAISS
Data Analysis & Visualization	Pandas, NumPy, Matplotlib, Seaborn, SciPy, MS Excel
Statistical Software	SPSS, Stata
Web & Deployment	FastAPI, Pydantic, Streamlit, Docker, Render, Railway, Vercel
Tools & Platforms	Git, GitHub, Kaggle

RESEARCH

Published

A Coverage Preserving Ensemble Framework with Minority Recovery for Robust Indoor Localization *
2026

- Developed a BLE-based indoor localization framework for nursing care environments using Random Forest classification, temporal smoothing, adaptive thresholding, and minority room recovery.
- Published in **International Journal of Activity and Behavior Computing**, Vol. 2026, Issue 2 (April 2026); presented at the **ABC 2026 Conference**.

Under Review

A Dual Pipeline Machine Learning Framework for Automated Multi Class Sleep Disorder Screening Using Hybrid Resampling and Ensemble Learning *
Jan 2026

- Designed a dual-pipeline architecture with SMOTETomek resampling to model physiological dependencies and address class imbalance.
- **Accepted** at *Social Network Analysis and Mining* (Q1 journal); achieved **98.67% accuracy**, validated using the Wilcoxon Signed-Rank Test.

Explainable Machine Learning for Chronic Kidney Disease Classification from Routine Urinalysis and Diabetes Records *
Sep 2026

- Developed interpretable ML models using grouped nested cross-validation to prevent patient-level leakage, with SHAP/LIME feature attribution and probability calibration.
- Urine measurements alone achieved **ROC AUC 0.973**; full model reached **0.964** (95% CI 0.942–0.982).

ONGOING RESEARCH

Recent Natural Hazard Exposure and Foundational Reading and Numeracy Skills among Children Aged 7–14 Years in Bangladesh Ongoing

- Nationally representative study using **MICS 2025** data examining the association between recent natural hazard exposure and foundational reading and numeracy outcomes among Bangladeshi children aged 7–14.

SYMTOLOGY: Generating a Code-Mixed Synthetic Dataset for Mental Health with an Ontology-Driven Prompting Framework in a Low-Resource Language Ongoing

- Building an ontology-driven prompting framework to synthesize code-mixed mental health data in a low-resource language setting.

Sensitivity of Short-Term CGM Forecasting to Evaluation Protocol, Historical Window Length, and Prediction Horizon Ongoing

- Systematic sensitivity analysis of short-term continuous glucose monitoring (CGM) forecasting models across evaluation protocols, historical window lengths, and prediction horizons.

Cross-Population Generalization in Short-Term Continuous Glucose Monitoring Forecasting: A Leakage-Controlled Evaluation Framework Ongoing

- Developing a leakage-controlled evaluation framework to assess cross-population generalization of short-term CGM forecasting models.

Three-Dataset Harmonization Strategy for Cross-Dataset Chronic Kidney Disease Analysis, External Validation, and Explainable Machine Learning Ongoing

- Harmonizing three CKD datasets for cross-dataset analysis, external validation, and explainable machine learning.

COMPETITIONS

Shobdotori: Where Dialects Flow into Bangla 2025

- Team **Backprop Sust** secured **4th place among 60+ teams**. Built and led the dual-stage Whisper+LoRA fine-tuning pipeline and BERT ensemble for punctuation restoration. Hosted by **CUET ETE**.

Build With Gemma @Bangladesh 2026

- Team **Backprop Sust** secured **4th place in the Online Track** by building **Khata**, a Gemma-powered application that transforms whiteboard photos into structured board-exam-ready practice material. Hosted by **Kaggle**, sponsored by **Google DeepMind**.

DUET AI Hackathon: RoadVision 2026

- Team **Backprop Sust** finished in the **top 15 of 101 teams** with a YOLOv8 vehicle detection pipeline achieving **mAP@0.50 of 0.623**. Hosted by **DUET CSE**, sponsored by **ICT Division Bangladesh**.

PoliMemeDecode — National Datathon 2025

- Team **Backprop Sust** ranked **20th out of 151 teams** using a **DeBERTa-v3**+**RoBERTa** dual-encoder with YOLO logo detection and InsightFace for multimodal meme classification. Hosted by **CUET CSE**.

PROJECTS

ScholarLink — Faculty Finder: AI-Powered Research Matchmaking Platform 2025

- End-to-end semantic search platform enabling students to discover faculty and PhD researchers using natural language queries at SUST (611+ faculty profiles indexed).
- Developed full-stack application using **React 18 + Vite** (frontend), **FastAPI** (backend), **MySQL**, Hugging Face embeddings, and **DeepSeek LLM** with server-sent events for streaming responses; containerized deployment with **Docker** on Railway and Vercel.
- **Links:** Live Demo | GitHub

Blood Glucose Prediction on OhioT1DM: Ensemble ML with Full XAI 2026

- Built a 30-minute-ahead CGM forecasting pipeline on the OhioT1DM benchmark using a six-algorithm ensemble and two LightGBM strategies; best ensemble achieved **RMSE 18.88 mg/dL** and **Clarke Error Grid A+B > 98.6%**.

- Integrated **SHAP** and **LIME** explainability across all pipelines, identifying `cgm_last`, `roc_5min`, and `cgm_recent_me` as dominant predictors.
- **Links:** GitHub *

Bengali LLM Hallucination Detection * 2026

- Built a cascaded multi-layer pipeline for classifying faithful vs. hallucinated Bengali LLM outputs, combining TF-IDF and BGE-M3 hybrid retrieval with context-span verification.
- Integrated a **Qwen2.5-32B-AWQ** judge with route-specific prompts and self-consistency voting to improve robustness across question types.
- **Links:** GitHub *

Bengali Long-Form Speech Recognition & Speaker Diarization * 2026

- Fine-tuned `whisper-medium` with iterative LoRA adapter stacking for long-form Bengali ASR, using chunked 30-second inference with overlap for hour-long audio.
- Built a speaker diarization pipeline using **ECAPA-TDNN** embeddings with ArcFace loss, BIC-based speaker counting, and Silero VAD; competed in both DL Sprint 4.0 tracks.
- **Links:** GitHub *

Roadside Object Detection * 2024

- Implemented real-time roadside object detection using **YOLOv11n** with a custom training pipeline; fine-tuned on a combined dataset (COCO + custom road imagery) with advanced augmentations (Mosaic, MixUp, HSV shifts).
- Achieved competitive mAP@0.5 scores and exported the model to ONNX/TensorRT for edge-device inference.
- **Links:** GitHub *

MENTORSHIP

Junior Python Mentor Jul 2025 – Sep 2025

SUST Data Science Club Sylhet, Bangladesh *(in-person)*

- Delivered in-person instruction for the 7-day **"PyNEXT"** course, guiding a cohort of **50 students** through Python fundamentals and troubleshooting challenges.
- Provided one-on-one support to strengthen participants' programming skills, fostering an engaging and collaborative learning environment.